

An MS/MS shortlist is not an identification: choosing a formula-free ranker and knowing when to decline

Study repository [rewire-bio/msms-model-selection](#)

Author list to be confirmed by the study owner. Drafted by an AI model; see Section 8.

Original analysis: 3–5 October 2026. Manuscript compiled from the archived results: October 2026.

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Abstract

A laboratory that can confirm five candidate structures for an MS/MS spectrum needs a formula-free method to choose those five from a mass-matched candidate list. On MassSpecGym formula split seed 1 as processed for MSAlign (10,648 test spectra from 1,421 molecules, $[M+H]^+$ and $[M+Na]^+$ only, identity judged on 2D structure), under a protocol frozen before any test score existed, two newly trained alignment models put the true structure in the top five for 63.2% (Emb-Cos, 95% interval [58.4, 67.5]) and 61.4% [56.2, 66.3] (DreaMS + Morgan) of test spectra on deduplicated official pools (median 252 structures), against 31.3% for a DeepSets fingerprint predictor, 13.3% for ranking by precursor mass error and 2.1% for random ordering. The pre-declared contrast between the two alignment models, -1.85 points $[-5.19, +1.35]$, showed no detectable difference, which is not equivalence. Recall@5 fell as pools grew (to 41.3% and 45.9% at a median of 449 structures), and the tie and duplicate convention moved mass error most (6.3% to 13.3%). In a target-removal stress test, a decline threshold fitted on validation for at most 10% false nominations gave DreaMS + Morgan shortlists for 21.3% of target-present test queries while still nominating for 13.2% of target-removed ones: the top score is only a weak signal that the answer is missing. A post hoc audit suggests that about half of mass error’s hits come from the 21% of test spectra whose precursor values look recomputed from the annotated structure. Retrained with released code and scored by its strict rule, DreaMS + Morgan reached 32.6% Recall@1, just below the published two-SD band; Emb-Cos fell below the band at all three cut-offs and DeepSets far above it. One seed is not a refutation of the published results. All values are existing results from one split seed and one training seed per model; independent reproduction is pending.

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1 Introduction

A tandem mass spectrum (MS/MS) records the fragments of one selected ion. Turning it into a molecular structure usually means ranking a list of candidate structures whose masses match the precursor, and then confirming the best few by orthogonal means. If a laboratory can afford to confirm five candidates, the quantity that matters is whether the true structure is among the first five ranked, Recall@5. This manuscript asks which formula-free ranking method should produce those five, and when a method should decline to nominate at all.

MSAlign v2 reports Recall@5 of 60.4–71.3% for its standalone alignment models on the MassSpecGym formula split (Table 1) [1]. Published numbers, however, depend on the split seed, the candidate-pool construction, the treatment of repeated structures and tied scores, and whether the true formula is known; they also assume the answer is in every pool. We retrained the formula-free models we could run with released code and settings, reproduced the released evaluation, and then measured them under a protocol frozen before any test score existed. The contributions are: (i) a reproduction check against published two-SD bands; (ii) a primary comparison of six formula-free rankers on deduplicated pools with molecule-grouped intervals and three pre-declared contrasts; (iii) stress tests of pool size and of tie and duplicate conventions; (iv) a controlled target-removal test of validation-fitted decline thresholds; (v) a precursor-error audit; and (vi) a companion tool that applies the same checks to a user’s own spectra. A shortlist is a set of structures to confirm, not an identification.

2 Background and related work

2.1 The task and the benchmark

The precursor m/z , adduct and charge give the neutral mass, and the structures whose exact masses fall within a tolerance of it, in parts per million (ppm), form the candidate pool. $[M+H]^+$ is the molecule plus a proton, $[M+Na]^+$ the molecule plus a sodium ion. In MassSpecGym, each pool holds up to 256 structures within 10 ppm of the annotated structure’s mass, drawn from three databases in priority order “until the maximum number of candidates $|C| = 256$ is reached” [2]. The pools are centred on the annotated structure’s mass, not on the measured precursor, so the true structure is in every pool by construction; a real search has to centre on the measured precursor. Many decoys are isomers with the same formula and exact mass: in the validation pools, 32.6% of distinct decoys had exactly the target’s monoisotopic mass. No mass rule can separate them; Kind and Fiehn showed that accurate mass cannot fix a unique formula even at 0.1 ppm [3].

Identity here is 2D: structures that differ only in stereochemistry, which the first block of an InChIKey leaves out, count as the same answer, and structures are compared as RDKit canonical SMILES with stereochemistry removed. Methods given the target’s true formula form a separate track, which MSAlign v2 reports in its own panel [1]; everything here is formula-free.

2.2 Learned ranking methods

A Morgan fingerprint is a bit vector of the circular substructures around each atom [4]. Contrastive alignment trains a spectrum encoder and a structure encoder so that a spectrum lands close to its true structure in a shared space and away from decoys; Emb-Cos is De Waele et al.’s name for this cosine formulation [5]. DreaMS is a transformer “pre-trained in a self-supervised way on millions of unannotated tandem mass spectra” [6]. MSAlign combines frozen spectrum and molecule encoders through small trained networks, a model zoo of representation pairs plus a Gaussian mass scorer, and late fusion calibrated on validation data (Figure 1) [1].

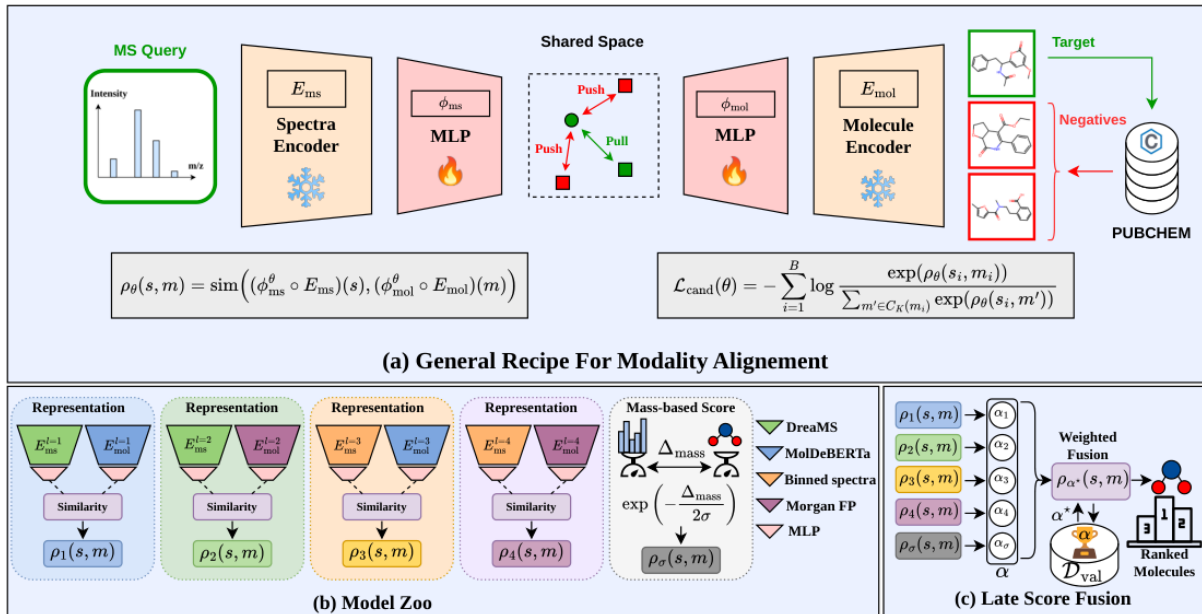


Figure 1: PUBLISHED. **The MSAlign recipe:** frozen encoders joined by small trained networks, a “model zoo” of representation pairs plus a Gaussian mass scorer, and late fusion calibrated on validation data. Of the representation pairs in panel (b), only DreaMS + Morgan fingerprint was retrained here; the MolDeBERTa pairs and the fused model are published numbers only. Figure 1 from Krzakala et al., arXiv:2605.19752v2 (2026) [1], CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>); cropped, caption removed.

2.3 Published numbers and their limits

MSAlign v2, posted on 25 September 2026, reports formula-free results on the MassSpecGym formula split as means and standard deviations over three split seeds (Table 1). Three things limit what such a table can tell a reader. First, the paper’s two versions disagree: v1 (19 May 2026) printed DeepSets at 22.2% Recall@1 and the DreaMS + fingerprint pair at 48.5%, whereas v2 prints 4.7% and 38.6% for the same split; v1’s headline 53.8% used ChemBERTa rather than MolDeBERTa, so it is a different model from v2’s 47.5% [1, 7]. Second, the strongest published rows were not run here: both v2 MSAlign models use MolDeBERTa and have no released checkpoints (they “will be released upon publication” [1]); MolDeBERTa is licensed CC BY-NC-ND 4.0 [8], and the candidate embedding cache would take about 20 GB. JESTR, and the formula-informed FLARE and MVP, need a Linux CUDA stack. Third, an audit by MassSpecGym’s authors and others found “evaluation issues in at least 17 of 26 papers reporting MassSpecGym benchmark results in the first year of its adoption”, and showed that filtering a mass-based pool to the true formula alone lifts a random ranker’s Hit@1 (Recall@1) from 0.43% to 12.98% [9]. A published Recall@k describes one pool construction, one split and one tie rule.

2.4 Ties, pool size and declining

Conventions differ across the literature. MSAlign’s released code is strict [10], and Gupta et al. broke ties “by random permutation” [11]. Guo et al., writing about recommender systems, note that such a gap “follows mechanically from the tie block; it is not evidence that the score identified the relevant item” [12]. Small pools inflate weak methods’ scores: Giné et al. found that at 0.1 ppm “the random-model baseline was substantially higher due to the smaller candidate pools” [13]. Declining is not a new idea either. COSMIC found the margin between hit and runner-up, and the number of candidates, to be “highly important features” for confidence [14], and MS2Query filtered unreliable matches with a score threshold [15]. Jürgens et al. concluded that “when

Table 1: PUBLISHED. **MSAlign v2 formula-free results** (Tables 3 and 4 of [1]), MassSpecGym formula split, Recall@k in %, mean $\pm$ SD over three split seeds, official pools of up to 256 candidates including repeated structures. The paper says only that a fixed tie-breaking rule was used; the released code counts ties against the target [10]. The standalone (47.5) and fused (58.2) MSAlign rows are different models. Formula-known methods are omitted because they receive the true formula. Transcribed from the original article; not re-extracted from the paper.

Method (MSAlign v2, published)	R@1	R@5	R@20
DeepSet (Fourier features, 50 epochs)	4.7 $\pm$ 0.6	13.0 $\pm$ 1.3	27.5 $\pm$ 2.0
JESTR	13.9 $\pm$ 0.5	32.5 $\pm$ 0.8	57.0 $\pm$ 0.7
Emb-Cos	42.7 $\pm$ 3.0	65.2 $\pm$ 2.8	80.4 $\pm$ 1.6
Binned + Fingerprint pair (Table 4)	34.3 $\pm$ 1.3	56.2 $\pm$ 2.4	71.8 $\pm$ 1.3
DreaMS + Fingerprint pair (Table 4)	38.6 $\pm$ 2.8	60.4 $\pm$ 2.1	77.6 $\pm$ 0.8
MSAlign, DreaMS + MolDeBERTa, standalone	47.5 $\pm$ 3.4	71.3 $\pm$ 2.6	87.0 $\pm$ 1.1
MSAlign fused (7 scores, 3 of them mass scores)	58.2 $\pm$ 3.4	80.7 $\pm$ 3.0	92.8 $\pm$ 1.1

the majority of predictions are incorrect, any procedure with valid risk guarantees must abstain from making predictions for most instances”, and named target-absent pools as “an important open-world setting” [16]; their default evaluation uses formula-filtered pools that always contain the target. A conformal study found coverage falling below the nominal level when calibration and test data came from different molecular clusters [17]. Gupta et al. and Giné et al. varied candidate sets in other ways [11, 13]. The specific combination studied here (controlled target removal with validation-fitted thresholds and false-nomination rates for formula-free alignment models, nested pool sizes from one mass window, and tie-rule effects) was not found in a dated literature search on 4 October 2026.

3 Materials and methods

3.1 Protocol

The evaluation protocol was frozen on 4 October 2026 at 22:07:05Z (SHA-256 aa2b30b5...31b6), before any test-fold outcome of any method was computed or viewed; smoke runs used only training and validation rows. One amendment, A1 (about 23:00 UTC the same day, before any learned-model test score existed), added a descriptive window-coverage measure (Section 4.6). The frozen original, the amended text and the freeze receipt are archived.

3.2 Data, split and populations

The data are MassSpecGym as processed by the MSAlign authors (Zenodo 10.5281/zenodo.22830464, CC BY 4.0) [2, 18], split `formula_seed1` (MSAlign v2 formula split, seed 1): training 210,825 spectra / 26,087 molecules; validation 9,631 / 1,428; test 10,648 / 1,421. All 10,648 test spectra form the primary population, none excluded; every spectrum carries an $[M+H]^+$ or $[M+Na]^+$ adduct, charge +1 and a precursor m/z . Spectra are not independent: 34.7% of test rows match more than one MassSpecGym record with the same structure, adduct, precursor and ten strongest peaks, so uncertainty is computed by resampling molecules. A pre-declared secondary descriptive population, the official MassSpecGym `mces_1` test fold (17,556 spectra, 2,997 molecules), was scored with the cheap baselines only (Appendix Table 15).

3.3 Candidate pools

Pools were built once per target molecule and saved with hashes before scoring:

- `official_raw`: the released 256-candidate mass list as distributed, duplicates counted with their multiplicity (the published setting). Once stereochemistry is stripped, 85.4% of the

Table 2: The formula-free methods run here. All learned models were newly trained with released code and settings, one seed (42) each. Fusion is exploratory and not a recommendation.

Method	Spectrum input	Structure input	Score
Random ordering	none	none	random order (expected value)
Precursor mass error	precursor m/z, adduct, charge	exact mass	smallest absolute ppm error first
DeepSets fingerprint predictor	peak list	Morgan fingerprint	cosine between predicted and candidate fingerprint
Emb-Cos	binned peaks	Morgan fingerprint	cosine in a learned shared space
DreaMS + Morgan (MSAlign code)	DreaMS embedding	Morgan fingerprint	cosine in a learned shared space
Alignment + mass fusion (exploratory)	DreaMS embedding and precursor	Morgan fingerprint and exact mass	weighted sum of z-scored DreaMS + Morgan and mass-error scores; mass weight 0.55 chosen on validation Recall@5

28,936 official lists contain at least one repeated structure (mean 6.9 repeats, maximum 121).

- **official_dedup (primary):** `official_raw` with repeated 2D structures removed; median 252 distinct structures.
- **sub16, sub64:** the target plus 15 or 63 decoys drawn without replacement from `official_dedup`, seeded by the SHA-256 of the target SMILES.
- **expanded1024:** `official_dedup` plus decoys from the MassSpecGym 4M molecule set (a different database snapshot) within 10 ppm of the target’s exact mass, excluding the target and existing members, capped at 1,024 structures (test median 449, IQR 296–691; 5.1% of the 2,849 validation and test targets reach the cap).
- **absent:** `official_dedup` with the target removed, a controlled stress test, not a measured rate of real absence.

The target is in every pool except **absent**. Pool-size distributions are in Appendix Table 9.

3.4 Methods compared

Every method receives the peak list (top 100 peaks as released), precursor m/z, adduct (hence charge), collision energy where the released model uses it, and the candidate structures, and nothing else (Table 2). All learned models were newly trained with released code and settings, one seed (42) each, on one Apple M4 machine with 16 GB of memory: DeepSets with the released `model_zoo` configuration (50 epochs, checkpoint by validation loss); Emb-Cos with the released configuration (16,000 steps, checkpoint by validation Recall@1); and the MSAlign alignment with the released `massspecgym_formula` configuration and the molecule representation set to Morgan (radius 2, 4,096 bits), 30,000 steps, checkpoint by validation Recall@1 (selected at step 3,296). The fusion weight was chosen on validation Recall@5 over $w \in \{0, 0.05, \dots, 1\}$ (Appendix Table 13). No reference spectrum library search was run, because it would require library spectra for every candidate.

3.5 Metrics and tie rules

Recall@k is the share of test spectra whose true structure is among the first k ranked candidates; Recall@5 on `official_dedup` is primary, with Recall@1 and Recall@20 secondary. The primary

tie rule is the expected value under uniform random tie-breaking: for a target with b decoys scoring strictly higher and t decoys tied,

$$\text{Recall}@k = \min\left(1, \max\left(0, \frac{k-b}{t+1}\right)\right), \quad (1)$$

so a target tied with three decoys counts as the average over the four possible orders. The released strict rule, in which a tied decoy ranks ahead of the target, is reported alongside and used for comparison with published numbers. Every test spectrum is in the denominator; a method that fails on a spectrum scores a miss, and there were no failures.

3.6 Uncertainty and pre-declared contrasts

Intervals are 95% percentile intervals from 2,000 bootstrap resamples of target molecules (seed 20261004), resampling all spectra of a chosen molecule together; paired contrasts use the same resampled molecules for both methods. Three contrasts on Recall@5, `official_dedup`, test, were declared before scoring: C1, DreaMS + Morgan minus mass error; C2, DreaMS + Morgan minus Emb-Cos; C3, DeepSets minus mass error. Everything else is descriptive. With one split seed and one training seed, intervals reflect which molecules were sampled, not variation between splits or training runs.

3.7 Declining to nominate

Confidence is the top-1 score in the pool (cosine for the learned models; minus the smallest absolute ppm error for mass error); the margin between the top two scores is a secondary, exploratory confidence. Each validation spectrum is scored twice, with `official_dedup` (target present) and with `absent`. Two thresholds are fitted on validation only and then frozen: one that keeps 90% of target-present validation queries covered, and the lowest threshold at which at most 10% of target-removed validation queries are nominated. On test, "covered" is the share of target-present queries given a shortlist and "false" the share of target-removed queries still given one.

3.8 Reproduction check

For the three trained models, the released evaluation (strict ties, `official_raw`) on test was compared with the v2 paper’s formula-split mean ± 2 SD. The scoring harness had to reproduce the trainer’s own validation Recall@1 for the selected checkpoint to 0.1 percentage points before any test scoring, and it reproduced each released evaluation exactly on validation and test; for Emb-Cos and DeepSets the harness test values equal the released code’s own end-of-training evaluation (archived `released-result.json` receipts of runs T02 and T03) to 0.01 points.

3.9 Execution and provenance

All runs used one Apple M4 machine (10 cores, 16 GB, macOS) without CUDA. Emb-Cos trained in 3.0 hours (16,000 steps) and DreaMS + Morgan in 7.75 hours (30,000 steps, selected checkpoint at step 3,296), so the difference in training time partly reflects different step budgets. Run receipts were written by each run into its own immutable directory. The tables below are formatted from the archived outputs; every archive member used is digest-checked against the migration audit, and values that also appear in the original article are cross-checked against it (Appendix D). None of the runs was re-executed for this manuscript.

Table 3: REPRODUCTION. Released evaluation of the retrained models on MassSpecGym formula split seed 1, test fold, strict ties, official lists with duplicates kept, Recall@k in %, one training seed (42) per model, against the v2 paper’s mean $\pm$ 2 SD. Source: archived `analysis/metrics.csv` (pool `official_raw`, rule `strict`).

Model (selected checkpoint)	R@1	R@5	R@20	v2 mean $\pm$ 2 SD (R@1 / R@5 / R@20)	Verdict
DreaMS + Morgan pair, MSAlign code (step 3,296 of 30,000)	32.6	56.8	76.7	33.0–44.2 / 56.2–64.6 / 76.0–79.2	R@1 just below; R@5, R@20 inside
Emb-Cos, model_zoo (epoch 3)	35.4	59.3	76.8	36.7–48.7 / 59.6–70.8 / 77.2–83.6	all three below
DeepSets + Fourier, model_zoo (epoch 3)	13.1	27.9	46.9	3.5–5.9 / 10.4–15.6 / 23.5–31.5	all three far above

4 Results

4.1 Reproduction: at one seed only DreaMS + Morgan lands mostly inside the published band

Table 3 compares each retrained model, scored by the released rule, with the v2 paper’s mean $\pm$ 2 SD. DreaMS + Morgan sits just below the band on Recall@1 and inside it on Recall@5 and Recall@20; it peaked early (validation Recall@1 0.376 after two epochs), and the released rule keeps that checkpoint. Emb-Cos fell below the band on all three cut-offs. DeepSets landed far above v2 on all three; at Recall@5 and Recall@20 it sits closer to v1’s printed 36.1 and 50.9, and at Recall@1 (13.1) roughly midway between v2’s 4.7 and v1’s 22.2. With one split seed and one training run, a miss shows where this run landed; it does not show that the paper is wrong.

4.2 New measurement: both alignment models lead and were not separated

Table 4 and Figure 2 give the primary results. Either alignment model put the true structure among the five confirmations for about six spectra in ten; mass error did so for fewer than one in seven. Every method scored every spectrum. The pre-declared contrasts (Table 5) show that C1 and C3 are well clear of zero: DreaMS + Morgan beat mass error by 48.0 points of Recall@5 and DeepSets beat it by 18.0. No contrast was declared between alignment and fingerprint prediction; the alignment models’ intervals ([56.2, 66.3] and [58.4, 67.5]) do not overlap the DeepSets interval [27.2, 35.5]. C2, at -1.85 points $[-5.19, +1.35]$, includes zero: no detectable difference between the two trained alignment models. That is not equivalence, and a modest gap in either direction is not ruled out. Nor is C2 a test of spectral pretraining: the models differ in spectrum input, architecture, loss settings, training budget and selected epoch, and each was trained once. The paper’s controlled pair, DreaMS + fingerprint at 38.6% against binned + fingerprint at 34.3% Recall@1, is published and was not reproduced here. The fusion with mass error reached 64.8% Recall@5, but it is exploratory, leans on recomputed precursor values (Section 4.6) and is not the recommendation.

4.3 More candidates mean fewer hits for every method

The pool-size stress test kept the target in every pool (Figure 3; Appendix Table 10). DreaMS + Morgan Recall@5 fell from 94.5% at 16 structures to 81.4% at 64, 61.4% at 252 and 41.3% at 449, and its Recall@1 from 69.2% to 19.1%. Random ordering fell from 31.2% to 1.3%. The order of method families held at every size: alignment models (with or without fusion), then DeepSets, mass error and random ordering. Within the top family it did not: DreaMS + Morgan led Emb-Cos at 16 structures (94.5% against 93.8%) and trailed at 449 (41.3% against 45.9%), where fusion (43.9%) also fell below Emb-Cos. The expanded-pool paired difference, -4.56 points $[-7.95, -1.19]$, was not pre-declared and uses decoys drawn differently from the official ones, so it

Table 4: NEW MEASUREMENT. **Primary results:** MassSpecGym formula split seed 1, 10,648 test spectra from 1,421 molecules, deduplicated official pools (median 252 structures), random-tie expectation, Recall@k in % with 95% molecule-bootstrap intervals (2,000 resamples). Random ordering is the mean of 100 random orderings (equal to the analytic expectation at printed precision), shown without an interval. Source: archived `analysis/metrics.csv` and `analysis/random-repeats.json`.

Method	R@1 [95% CI]	R@5 [95% CI]	R@20 [95% CI]
Random ordering (expectation)	0.4	2.1	8.5
Precursor mass error	3.7 [2.9, 4.5]	13.3 [11.2, 15.6]	32.0 [28.4, 35.5]
DeepSets fingerprint predictor	13.4 [10.6, 16.5]	31.3 [27.2, 35.5]	53.9 [49.4, 57.9]
Emb-Cos	36.1 [31.6, 40.9]	63.2 [58.4, 67.5]	82.4 [79.5, 85.1]
DreaMS + Morgan alignment	33.2 [28.3, 38.4]	61.4 [56.2, 66.3]	83.9 [80.8, 86.8]
Alignment + mass fusion (exploratory)	35.2 [30.4, 40.2]	64.8 [59.6, 69.9]	86.2 [83.0, 89.0]

Table 5: NEW MEASUREMENT. **Pre-declared contrasts** (Recall@5, deduplicated official pools, test fold): paired differences in percentage points on the same bootstrap resamples of molecules. The secondary contrasts at other cut-offs, pools and rules are descriptive (Appendix Table 16).

Pre-declared contrast (Recall@5)	Difference, points [95% CI]
C1: DreaMS + Morgan minus mass error	+48.0 [+42.2, +53.6]
C2: DreaMS + Morgan minus Emb-Cos	-1.85 [-5.19, +1.35]
C3: DeepSets minus mass error	+18.0 [+13.4, +22.5]

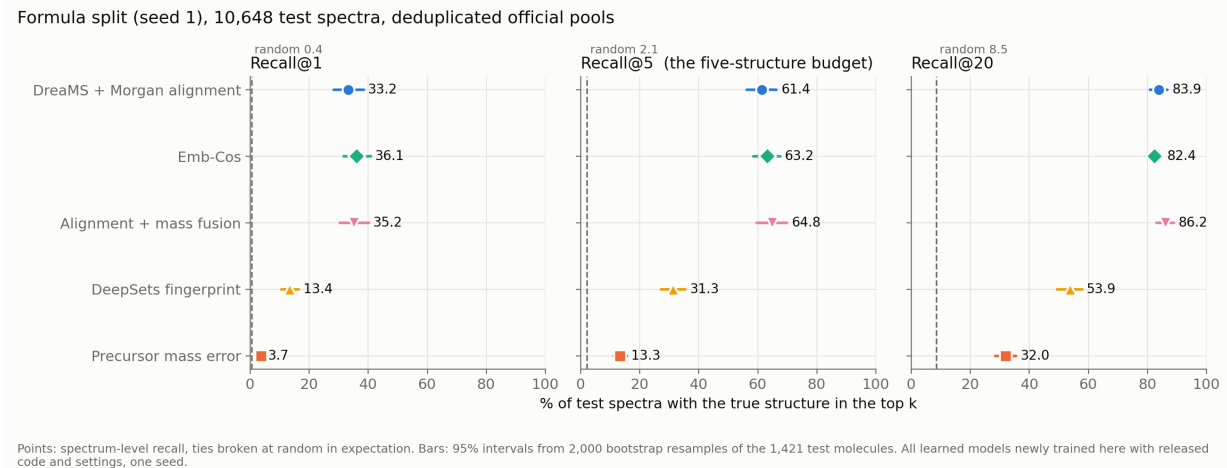


Figure 2: NEW MEASUREMENT. **Recall@1, @5 and @20 by method** on MassSpecGym formula split seed 1 (10,648 test spectra, deduplicated official pools), ties broken at random in expectation. Bars are 95% intervals from 2,000 bootstrap resamples of the 1,421 test molecules; dashed lines are the random expectation. Higher is better. Original figure `article/assets/01-recall-by-method.png`, drawn by the companion’s `make_charts.py` from the archived analysis.

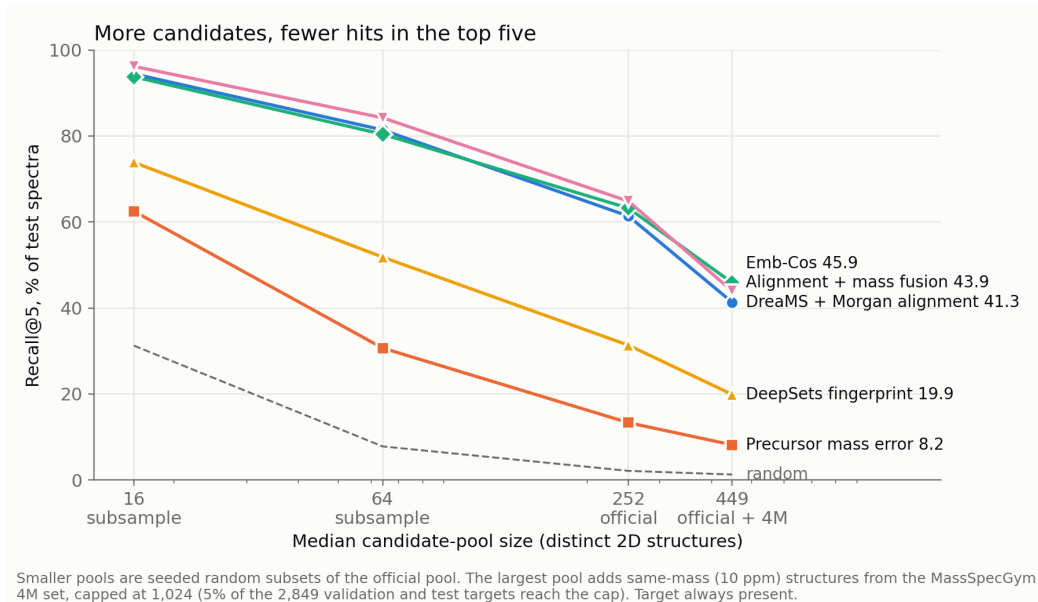


Figure 3: NEW MEASUREMENT (stress test). **Recall@5 against median pool size**, random-tie expectation, target always present, 10,648 test spectra. The 16 and 64 pools are seeded subsets of the official pool; the 449 pool adds same-mass structures from the MassSpecGym 4M set, a different database snapshot. Pool size is on a log scale. Higher is better. Original figure [article/assets/02-pool-size-stress.png](#).

Table 6: NEW MEASUREMENT. **Recall@5 / Recall@1 (%)** under four counting conventions applied to the same test scores. The first column is the released evaluation; the last is this study’s primary rule. Source: archived [analysis/metrics.csv](#).

Method	Strict, duplicates kept	Strict, duplicates removed	Random-tie, duplicates kept	Random-tie, duplicates removed
Mass error	6.3 / 0.8	6.3 / 0.8	12.9 / 3.6	13.3 / 3.7
DeepSets	27.9 / 13.1	30.3 / 13.1	28.9 / 13.3	31.3 / 13.4
Emb-Cos	59.3 / 35.4	62.4 / 35.4	60.4 / 36.0	63.2 / 36.1
DreaMS + Morgan	56.8 / 32.6	60.5 / 32.6	57.9 / 33.1	61.4 / 33.2
Fusion (expl.)	60.1 / 34.6	63.9 / 34.6	61.2 / 35.0	64.8 / 35.2

is a reason to test both models at one’s own list size, not a ranking. An expected Recall@5 should be quoted with its list size.

4.4 Scoring conventions move the cheap baseline most

Figure 4 and Table 6 apply counting conventions for ties and duplicates to identical scores. Mass error doubled, from 6.3% Recall@5 under the strict rule with duplicates kept to 13.3% under the primary rule, and its Recall@1 rose from 0.8% to 3.7%. The learned models moved by a few points: DreaMS + Morgan from 56.8% to 61.4%. Mass error is hit hardest because every isomer of the target ties with it, and the strict rule counts all of them ahead of the target. MSAlign v2’s single Gaussian mass scorers reach only 0.6–0.8% Recall@1 on their own [1], which is consistent with strict counting on isomer-rich pools; that link is an inference. The rule should be stated next to any reported number.

4.5 The top score is only a weak signal that the answer is missing

Benchmark pools always contain the answer; a real search may not. Table 7 and Figure 5 report the target-removal stress test. Under the at-most-10%-false rule, DreaMS + Morgan nominated for 21.3% of target-present test queries (68.4% of those shortlists held the answer) and still

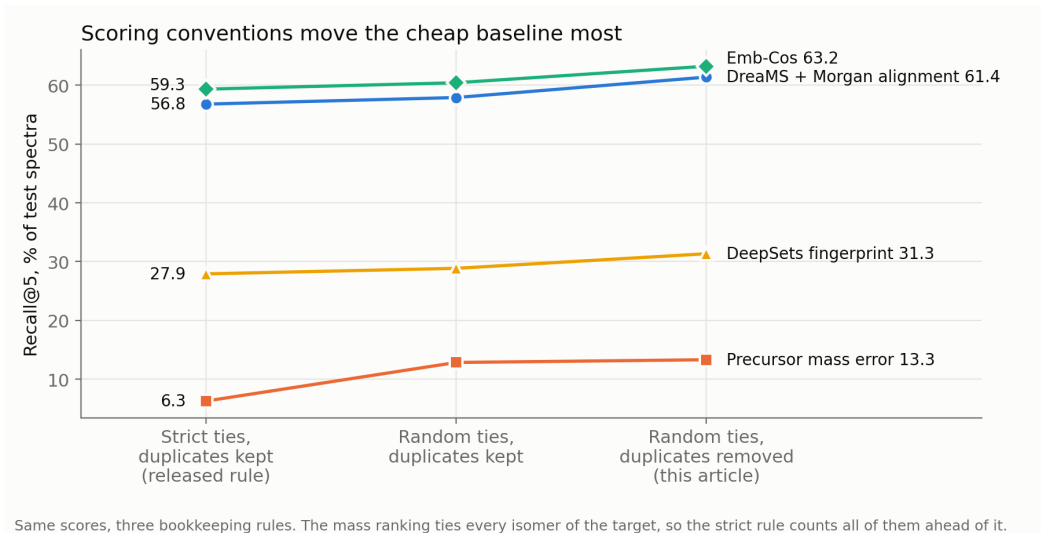


Figure 4: NEW MEASUREMENT. **Recall@5** under three bookkeeping rules applied to the same scores: the released strict rule with duplicates kept, random-tie expectation with duplicates kept, and random-tie expectation with duplicates removed (the primary rule). Original figure [article/assets/03-tie-and-duplicate-rules.png](#).

Table 7: NEW MEASUREMENT (stress test, artificial target removal). **Validation-fitted decline thresholds applied once to test**, deduplicated official pools, confidence = top-1 score. “Covered” is the share of target-present test queries given a shortlist; “false” the share of target-removed test queries still given one; 95% molecule-bootstrap intervals. Source: archived [analysis/abstention.csv](#).

Method	Threshold rule (fitted on validation)	Covered [95% CI]	R@5 among covered	False nominations [95% CI]
DreaMS + Morgan	at most 10% false	21.3% [16.2, 26.6]	68.4%	13.2% [9.0, 18.0]
DreaMS + Morgan	keep 90% covered	90.8% [88.5, 92.8]	64.3%	86.6% [83.2, 89.4]
Emb-Cos	at most 10% false	25.5% [22.0, 29.1]	83.7%	15.7% [12.5, 19.1]
Emb-Cos	keep 90% covered	90.8% [89.1, 92.2]	66.6%	85.6% [83.3, 87.7]
DeepSets	at most 10% false	18.4% [14.1, 23.1]	52.8%	15.1% [11.0, 19.5]
Fusion (expl.)	at most 10% false	21.8% [18.0, 26.0]	86.8%	10.2% [7.7, 13.0]
Mass error	at most 10% false	12.2% [9.9, 14.7]	22.1%	12.1% [9.8, 14.6]

nominated for 13.2% of target-removed ones; its validation-fitted threshold is 0.5994. The test false-nomination rates of the three single learned models (13.2–15.7%) all exceeded the 10% validation target. Using the margin between the top two scores instead kept false nominations under target but covered fewer queries: 13.8% covered with 9.6% false for DreaMS + Morgan, and 18.4% with 9.8% for Emb-Cos (Appendix Table 11). Mass error has no signal here: its coverage (12.2%) equals its false-nomination rate (12.1%), because removing the target leaves its isomers with the same top score. The consequence is a choice of failure mode. The 10% threshold declines about four in five target-present queries; the 90%-coverage threshold nominates for 86.6% of target-removed ones. In neither case is a cosine score a probability that the top structure is right. Figure 6 reproduces the selective-prediction framing of Jürgens et al. [16].

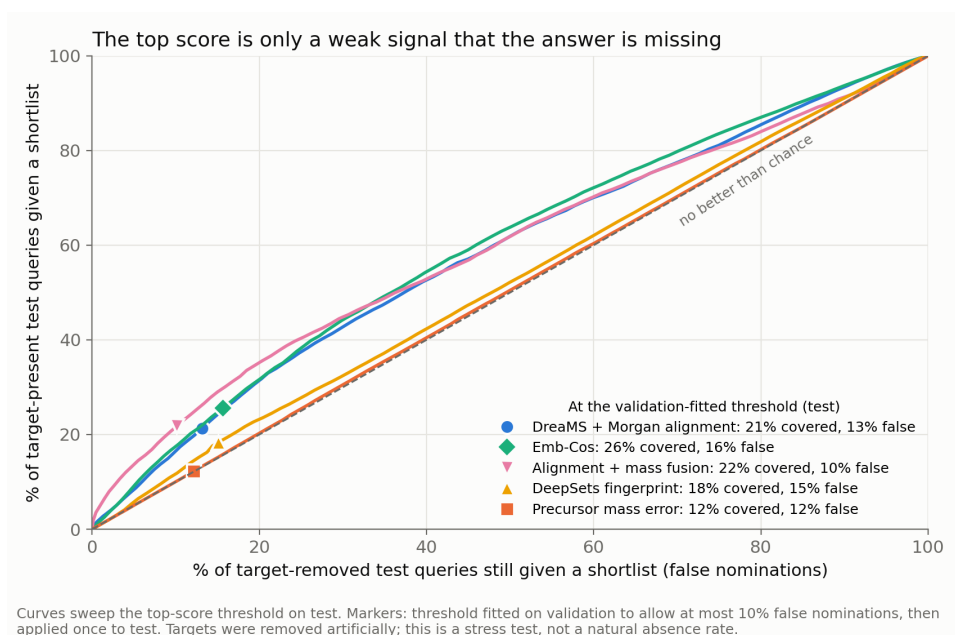


Figure 5: NEW MEASUREMENT (stress test, artificial target removal). **Coverage of target-present queries against false nominations on target-removed queries**, sweeping the top-score threshold on test; markers show the threshold fitted on validation for at most 10% false nominations. The diagonal is no better than chance. Legend values are rounded to whole percentages; Table 7 gives one decimal. Original figure [article/assets/04-decline-to-nominate.png](#).

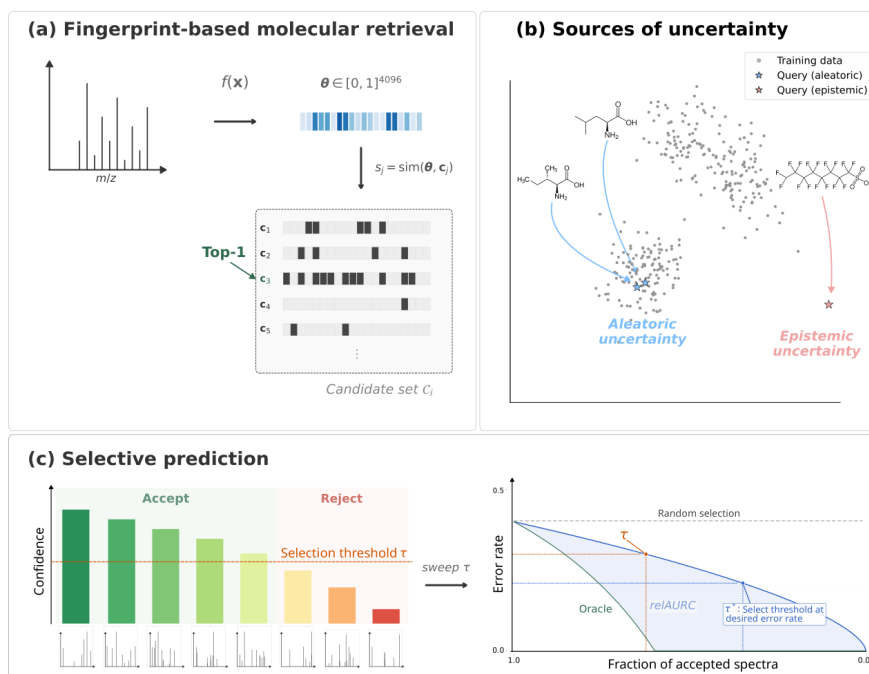


Figure 6: PUBLISHED. **Selective prediction as framed by Jürgens et al.:** retrieval with a top-1 pick, sources of uncertainty, and accepting or rejecting by a confidence threshold traced as a risk-coverage curve. Their default evaluation uses formula-filtered pools that always contain the target, unlike the target-removal test here. Fig. 1 from Jürgens et al., arXiv:2603.10950v2 (2026) [16], CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>); cropped, caption removed.

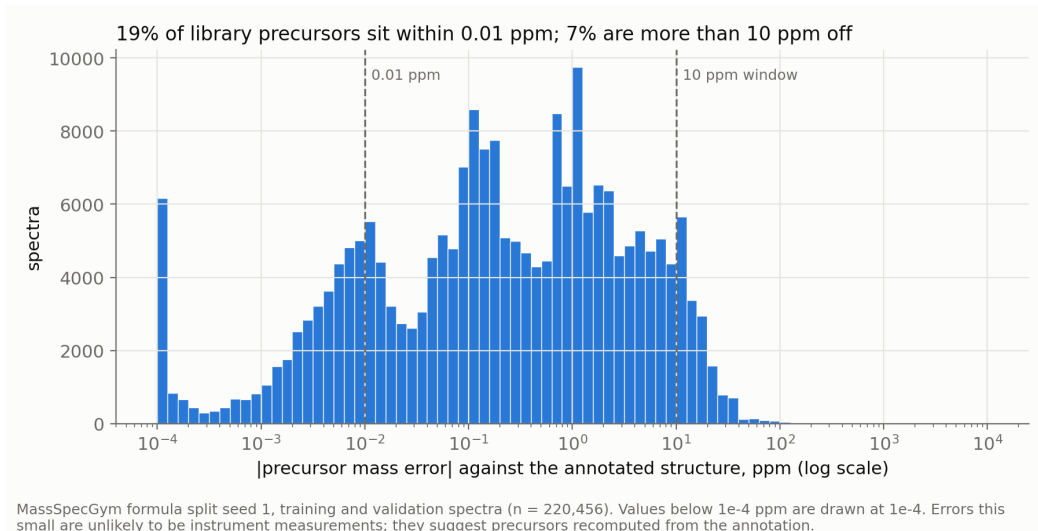


Figure 7: NEW MEASUREMENT (data audit). **Absolute precursor mass error against the annotated structure** for training and validation spectra ($n = 220,456$): 19.1% within 0.01 ppm, 7.07% more than 10 ppm away, median 0.211 ppm. The chart title rounds these to 19% and 7%. Values below 10^{-4} ppm are drawn at 10^{-4} . Original figure [article/assets/05-precursor-mass-error.png](#); per-fold summaries are in Appendix Table 14.

Table 8: POST HOC, descriptive. **Recall@5 (%) by precursor-error stratum**, test fold, deduplicated official pools, random-tie expectation, with 95% molecule-bootstrap intervals. Source: archived [analysis/strata-post-hoc.csv](#).

Stratum (post hoc)	Spectra / Mass molecules error	DeepSets	DreaMS + Morgan	Emb-Cos	Fusion	
Within 0.01 ppm (recomputed-looking)	2,264 / 526	31.6 [27.5, 35.8]	23.3 [17.9, 29.4]	54.3 [47.9, 60.3]	59.0 [54.1, 63.6]	62.7 [56.7, 68.2]
0.01–10 ppm	7,645 / 1,002	9.2 [7.2, 11.7]	34.0 [29.2, 38.8]	61.8 [55.7, 68.0]	63.7 [58.1, 69.0]	64.2 [58.2, 70.5]
Beyond 10 ppm	739 / 105	0.1 [0.0, 0.2]	28.3 [16.9, 39.2]	78.5 [61.0, 89.5]	71.2 [54.4, 82.7]	77.4 [59.4, 88.6]

4.6 Check the precursor value before trusting any ranking

In the training fold ($n = 210,825$), 19.0% of spectra sit within 0.01 ppm of the annotated structure’s theoretical mass (Figure 7; Appendix Table 14). Errors below 0.01 ppm are far tighter than the high (under 5 ppm) or very high (under 1 ppm) mass accuracy that Kind and Fiehn describe [3]; they suggest values recomputed from the annotation, as library curation can do: Spectraverse documents that “parent masses were recalculated from SMILES” [19]. At the other end (amendment A1, exploratory), 6.94% of test spectra lie more than 10 ppm from the annotated structure and 16.1% more than 5 ppm (validation 5.43% and 9.2%; `mces_1` test 4.18% and 15.4%). These spectra keep their answer only because the benchmark pools are centred on the annotated structure’s mass; a real search centred on the measured precursor would lose it before any ranking, so for these spectra the benchmark overstates what a real search would recover.

A post hoc split of the test fold by precursor error (Table 8) shows where the mass baseline gets its strength: 31.6% Recall@5 on recomputed-looking precursors and 9.2% on the 0.01–10 ppm group, which looks more like measured data. The 21% of test spectra in the first group supply about half of all its hits. Fusion added 8.4 points over DreaMS + Morgan on recomputed precursors and 2.4 points on the 0.01–10 ppm group, so much of its benefit comes from precursor values that real measurements would not provide, which is why fusion stays exploratory. Mass is still worth using as a filter, because it defines the pool; it should not be the ranker.

4.7 The companion tool on the declared demonstration spectra

Inputs and outputs. Spectra go in an MGF file with PEPMASS, CHARGE and ADDUCT in every block (COLLISION_ENERGY optional); candidates go in a CSV with `id` and `smiles` columns, frozen before any scores are looked at. Supported adducts are $[M+H]^+$, $[M+Na]^+$, $[M+NH_4]^+$, $[M+K]^+$ and $[M-H]^-$; the trained model saw only the first two, and other adducts run with a warning. Without `-model`, candidates are ranked by mass error alone and the run never declines on score. With the model, the tool declines when the top score is below the value in `models/thresholds.json`: 0.5994, the lowest value that nominated at most 10% of target-removed validation queries. The ranked CSV, still written after a decline, carries the score, `ppm_error`, a decision column and tie flags (`tied_with_previous`), and a provenance file is written alongside it that records the ppm window used.

Demonstration. The companion tool follows the path in Figure 8: check metadata, check the ppm window, rank, compare the top score with the validation-fitted threshold, then nominate five or decline. Mass-only runs skip the threshold step and never decline on score. Its executed notebook ran the tool on two MassSpecGym test spectra chosen by a fixed rule before their scores were viewed (the first $[M+H]^+$ and the first $[M+Na]^+$), and ran the $[M+H]^+$ spectrum again with its true structure removed from the candidate file (verbatim outputs in Appendix B). In the mass-only run on MassSpecGymID0226249 ($[M+H]^+$, Orbitrap, 254 candidates), the annotated structure is one of 30 candidates tied at the same mass, so the five nominated are an arbitrary cut. The model ranks it first yet declines, because its top score, 0.019, is below the 0.599 threshold; with the structure removed it declines too. The $[M+Na]^+$ spectrum MassSpecGymID0395778 (QTOF) fails before ranking: its recorded precursor, 826.4, is 117 ppm below the ion implied by the annotated structure, so no candidate lies within 10 or 50 ppm; only the 200 ppm window readmits 227 of the file’s 229 candidates, and the model then declines. Two deliberately invalid blocks were refused (missing adduct; charge inconsistent with the adduct). The DreaMS weights reproduced the released embeddings (minimum cosine 0.99999988 over 64 spectra), and the exported model matched the benchmark harness on the demo spectrum to within 2.2×10^{-4} , with the same rank.

4.8 Reproduction status

Not yet reproduced under this repository’s harness. The values above are imported from the 3–5 October 2026 runs. The reproduction reported in Section 4.1 is the original study’s own re-run of released code, not an independent reproduction. The repository’s reproduction protocol has not been designed, approved or executed.

5 Discussion

5.1 What to do in each situation

If five candidate structures can be confirmed, a learned alignment model is the clear first choice for picking them on this benchmark. The two alignment models showed no detectable difference; DreaMS + Morgan is the choice for users of the companion tool as shipped (it is the bundled model), and the cheaper Emb-Cos recipe is reasonable for those training their own model, who will need to score with it and fit its own decline threshold. More specifically:

- **Before ranking, check the precursor.** Compute the ppm error between the recorded precursor and the best-supported candidate masses, and look at how many decimals the precursor carries. A value recorded to one decimal place, like the 826.4 above, is too coarse for a 10 ppm window at that mass. Recheck the raw file, adduct and charge first.
- **If no candidate lies within the window,** recheck the precursor rather than widening the window. In a frozen file a wider window only readmits that file’s candidates; against a

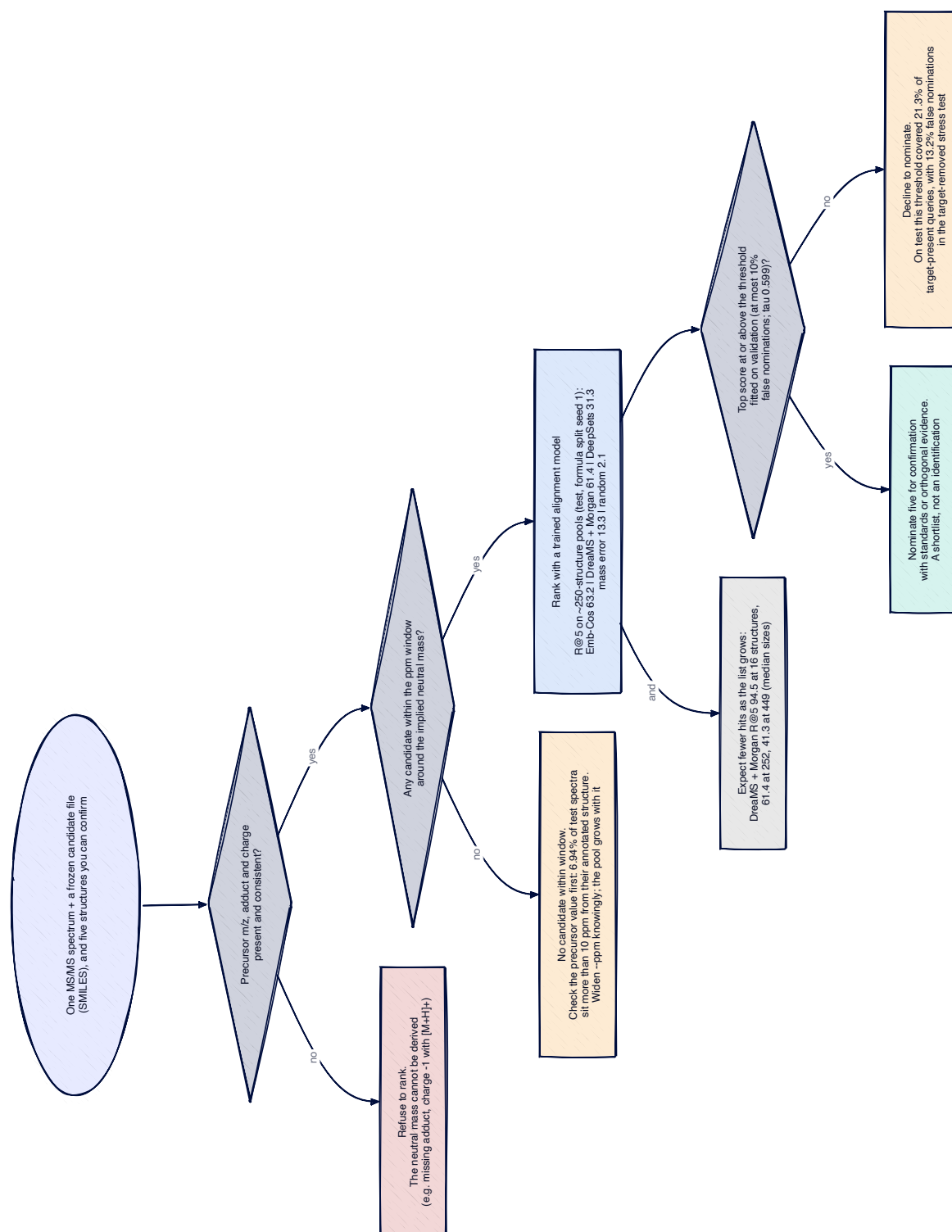


Figure 8: The decision path the companion tool follows for one spectrum, with the benchmark numbers that bear on each step (NEW MEASUREMENT and stress-test figures from Tables 4 and 7, the pool-size stress test and the precursor audit). Mass-only runs skip the threshold step and never decline on score. Converted from the original `article/assets/01-decision-path.svg` (D2 source `01-decision-path.d2`).

real database it admits far more structures, recall falls as the pool grows (Figure 3), and the shipped threshold was fitted on pools of about 250 mass-matched candidates. Report the window beside any shortlist.

- **After a decline**, the ranked list is still written, so a laboratory that prefers to inspect every list can ignore the decision. Treat a decline as triage, not as evidence of absence, since most target-present queries are declined too. With labelled spectra from one’s own instrument and candidate source, refit the threshold on them [17].
- **With mass-only ranking**, expect a shortlist every time; when a tie straddles the shortlist boundary, rank with an alignment model rather than confirm an arbitrary five.
- **Quote the list size** next to any expected hit rate, and if a list is much longer than 252 structures, expect less than the official-pool numbers.

5.2 What would change these conclusions

Three results would change this advice: further seeds that separate Emb-Cos from DreaMS + Morgan; released MolDeBERTa checkpoints showing whether the published standalone MSAlign’s 71.3% Recall@5 (v2, three-seed mean) holds under this protocol; and a collapse towards chance on generated decoys that resemble library compounds, or under spectrum masking, which would mean the alignment models lean on structure shortcuts.

6 Limitations

- **One split seed, one training seed.** The intervals reflect which molecules were sampled, not variation between splits or training runs.
- **Untested methods.** MSAlign v2 standalone and fused (both MolDeBERTa), JESTR, FLARE, MVP and reference-spectrum library search.
- **Untested data.** No other split seeds, MCES-split training, Spectraverse, NPLIB1, instrument-held-out evaluation, real target absence, negative-mode spectra or adducts beyond $[M+H]^+$ and $[M+Na]^+$. The post hoc instrument strata (pre-specified as descriptive; DreaMS + Morgan Recall@5 57.2% [51.3, 63.1] on Orbitrap, 71.2% [64.7, 76.8] on QTOF; Appendix Table 12) are confounded by which molecules each instrument measured.
- **Possible encoder overlap.** The DreaMS checkpoint loaded by the MSAlign code was fine-tuned on a MoNA subset (about 25,000 spectra, 5,500 InChI connectivity blocks) [6], and MassSpecGym draws spectra partly from MoNA [2]. Overlap with the test molecules was not checked: an unmeasured possibility, not a finding.
- **Near-duplicate records.** 34.7% of test rows match more than one MassSpecGym record with the same structure, adduct, precursor and ten strongest peaks, so spectra are not independent; that is why the intervals resample molecules.
- **Benchmark pools contain the answer.** Target removal is artificial, and pools centred on the annotated mass keep answers that a search centred on the measured precursor would lose (Section 4.6).
- **Shortcut controls.** The alignment models were not tested against generated decoys that resemble library compounds [11], nor by “masking all input MS/MS spectra with a constant value and confirming that performance degrades accordingly” [9].

- **Exploratory and post hoc analyses.** Fusion, the margin confidence, the expanded-pool contrast, the precursor and instrument strata and the window-coverage measure are exploratory or post hoc. Only C1–C3 were pre-declared contrasts; the instrument strata and the `mces_1` population were pre-declared as descriptive. None of the intervals is adjusted for multiplicity.
- **No identification.** The MSI minimum for a level 1 identification of a known metabolite is “a minimum of two independent and orthogonal data relative to an authentic compound analyzed under identical experimental conditions” [20]. The confidence scale of Schymanski et al. [21] invites researchers to define sublevels “on a per-study basis where evidence supporting different proposed structures is clearly presented” [22]. Every nominee should be confirmed against an authentic standard.

7 Data, code and licence availability

The study repository (`rewire-bio/msms-model-selection`) contains this manuscript and its build (`make paper-imported`); the original article (`article/original.md` and the byte-identical `article/published-original.md`, also embedded in this PDF as Supplement S1); the companion code (`companion/`, also as `downloads/msms-shortlist-companion-code.tar.gz`, 0.18 MB); the trained DreaMS + Morgan model and threshold (45 MB, split for a 25 MiB file limit; see below); and the results archive (`downloads/msms-shortlist-results.tar.gz`, about 7 MB) with analysis outputs, per-spectrum predictions and run receipts. The DreaMS TorchScript weights (468 MB, MIT licence [23]) are not bundled and are downloaded from Hugging Face `roman-bushuiev/DreaMS` at revision `c81a62766b10`.

The model bundle consists of five files in `downloads/`:

- `msms-shortlist-dreams-morgan-model.tar.gz.part01`, `.part02` and `.part03`;
- `reassemble-model.sh` and `msms-shortlist-dreams-morgan-model.parts.sha256`.

Saving all five in one folder and running `sh reassemble-model.sh` checks each part, joins them in order, verifies the SHA-256 of the joined archive and extracts the model. The joined archive’s SHA-256 is

```
646bbe4fbb14bda60e3544d4a963c9319b6564663c9b0752cd5e92b879eefc9.
```

The companion code is MIT-licensed, and `towers.py` adapts MSAlign inference code (MIT). Data and weights keep their own licences: MassSpecGym (MIT), the MSAlign Zenodo bundle (CC BY 4.0), DreaMS weights (MIT); MolDeBERTa (not used) is CC BY-NC-ND 4.0. The two reproduced literature figures are CC BY 4.0 and are attributed in their captions.

8 AI assistance, review and disclosure

This manuscript was drafted by an AI model (Claude Opus, `claude-opus-5-5`) from the archived evidence. A draft was reviewed by an independent AI reviewer instance of the same model (`evidence/paper-migration/review.md`, 18 findings), and the author’s disposition of every finding is in `evidence/paper-migration/review-disposition.md`; this version incorporates those corrections. The original article does not state who ran the analyses or how it was reviewed, and this manuscript makes no claim about either. No human scientific review of this manuscript has taken place.

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A Reproducing the historical run

The commands below are those recorded in the original article and the companion README; they were executed for the original article and were *not* re-executed for this manuscript. `uv sync` uses `uv.lock` (torch 2.2.1, rdkit 2023.9.6, numpy 1.25.0, pandas 2.2.1).

```
uv sync

# Mass-error ranking only (no model files needed)
uv run msms-shortlist --spectra examples/mh.mgf \
  --candidates examples/mh-candidates.csv --out out/mh-mass.csv

# Trained DreaMS + Morgan alignment model
uv run msms-shortlist --spectra examples/mh.mgf \
  --candidates examples/mh-candidates.csv \
  --model models/dreams_morgan_formula_seed1_fp16.pt \
  --dreams weights/DreaMS_embedding_model_torchscript.pt \
  --thresholds models/thresholds.json --out out/mh-model.csv
```

From a clean archive, `uv sync` took 1.5s with a warm cache, the first CLI run 35s including the package build, and a model run 5.3s. The benchmark pipeline (MSAlign checkout at `c2ef425`, one storage patch, training, frozen pools, scoring, rank statistics, fusion and analysis) is listed command by command in `companion/REPRODUCE.md`, with data hashes and run receipts.

Rebuilding this manuscript. `make paper-imported` runs `scripts/build_paper.py` with the system Python and local TeX Live. It verifies the archive and member digests, formats the tables in `paper/generated/`, converts the original decision-path SVG with `rsvg-convert`, and runs `pdflatex`, `BibTeX` and `pdflatex` twice. It performs no experiment, data download, model inference or environment creation. The receipt is `evidence/paper-migration/build-receipt.json`.

B Recorded demonstration outputs

Verbatim from the original article; each block is output from the run named above it, and lines without a query identifier were printed by the notebook from the run's provenance file or ranked CSV.

```
# Mass-only run, MassSpecGymID0226249 ([M+H]+, Orbitrap, 254 candidates)
MassSpecGymID0226249: nominate shortlist
tie at shortlist boundary: True | candidates sharing the 5th score: 30

# Model run, same spectrum
MassSpecGymID0226249: decline: top score below validation threshold
decision: decline: top score below validation threshold | top score: 0.0194 | threshold: 0.5994
rank of the annotated structure: 1

# Model run, true structure removed
MassSpecGymID0226249: decline: top score below validation threshold
decision with the true structure removed: decline: top score below validation threshold | top score:
-0.0245

# [M+Na]+ model runs, MassSpecGymID0395778 (QTOF), --ppm 10, 50 and 200
MassSpecGymID0395778: no candidate within window (no structure within 10.0 ppm)
10 ppm: no candidate within window | candidates in window: 0
MassSpecGymID0395778: no candidate within window (no structure within 50.0 ppm)
50 ppm: no candidate within window | candidates in window: 0
MassSpecGymID0395778: decline: top score below validation threshold
200 ppm: decline: top score below validation threshold | candidates in window: 227

# Invalid-metadata run (mass-only)
no_adduct_example: refused (missing ADDUCT; the neutral mass cannot be derived)
wrong_charge_example: refused (charge -1 is inconsistent with adduct [M+H]+)
```

The archived consistency check between the CLI and the benchmark harness on the demonstration spectrum (`receipts/X02-cli-demo-.../consistency-check.txt`) reads, verbatim:

candidates matched 252 of 254 | max abs diff 0.00021570734176634687 | Spearman 0.999993
 harness rank of target among dedup pool: 1 | CLI rank: 1

C Supplementary tables

All values are formatted from the archived results without recomputation.

Table 9: NEW MEASUREMENT, descriptive. **Candidate-pool sizes** per test spectrum (10,648 spectra), counting distinct 2D structures; for **official**, **raw** the multiplicity-weighted list length is not summarised in the archive, so its row equals the deduplicated one. Source: **pool-sizes.csv** in the archived **analysis/**.

Pool	Spectra	Mean	Min	25%	Median	75%	Max
absent	10,648	239.4	20	242	251	254	255
expanded	10,648	515.5	21	296	449	691	1024
official, dedup.	10,648	240.4	21	243	252	255	256
official, raw	10,648	240.4	21	243	252	255	256
16	10,648	16.0	16	16	16	16	16
64	10,648	64.0	21	64	64	64	64

Table 10: NEW MEASUREMENT (stress test). **Recall@5 and Recall@1 (%) by pool**, random-tie expectation, target always present; random ordering is the mean of 100 random orderings (equal to the analytic expectation at printed precision). Source: archived **analysis/metrics.csv** and **analysis/random-repeats.json**.

k	Method	Median pool size			
		16	64	252 (official)	449 (expanded)
R@5	Random	31.2	7.8	2.1	1.3
R@5	Mass error	62.5	30.7	13.3	8.2
R@5	DeepSets	73.9	51.8	31.3	19.9
R@5	Emb-Cos	93.8	80.4	63.2	45.9
R@5	DreaMS + Morgan	94.5	81.4	61.4	41.3
R@5	Fusion (expl.)	96.2	84.2	64.8	43.9
R@1	Random	6.2	1.6	0.4	0.3
R@1	Mass error	23.4	10.6	3.7	1.9
R@1	DeepSets	41.0	24.0	13.4	7.6
R@1	Emb-Cos	69.4	53.9	36.1	23.3
R@1	DreaMS + Morgan	69.2	52.1	33.2	19.1
R@1	Fusion (expl.)	71.6	54.9	35.2	20.0

Table 11: NEW MEASUREMENT (stress test; margin-confidence rows exploratory). **All decline-threshold results:** both confidences (top-1 score and the exploratory top-two margin) and both rules, with the fitted threshold, its validation false-nomination rate (%) and test coverage, Recall@5 among covered and false nominations (% , 95% intervals). Source: archived `analysis/abstention.csv`.

Method	Conf.	Rule	Threshold	Val. false	Covered	R@5 among covered	False nominations
DreaMS + Morgan	top1	at most 10% false	0.5994	10.0	21.3 [16.2, 26.6]	68.4 [54.3, 82.3]	13.2 [9.0, 18.0]
DreaMS + Morgan	top1	keep 90% covered	0.0405	84.2	90.8 [88.5, 92.8]	64.3 [58.8, 69.4]	86.6 [83.2, 89.4]
DreaMS + Morgan	margin	at most 10% false	0.2713	10.0	13.8 [10.0, 18.6]	85.2 [77.0, 91.0]	9.6 [6.8, 13.0]
DreaMS + Morgan	margin	keep 90% covered	0.0086	89.4	88.8 [84.8, 91.9]	62.7 [57.4, 67.7]	88.6 [84.6, 91.7]
Emb-Cos	top1	at most 10% false	0.4653	10.0	25.5 [22.0, 29.1]	83.7 [75.7, 89.9]	15.7 [12.5, 19.1]
Emb-Cos	top1	keep 90% covered	0.2267	84.5	90.8 [89.1, 92.2]	66.6 [61.6, 71.1]	85.6 [83.3, 87.7]
Emb-Cos	margin	at most 10% false	0.1088	10.0	18.4 [15.2, 21.7]	89.2 [85.0, 92.3]	9.8 [7.9, 12.0]
Emb-Cos	margin	keep 90% covered	0.0046	88.2	87.0 [83.0, 90.2]	65.9 [61.5, 69.7]	85.8 [82.0, 88.9]
DeepSets	top1	at most 10% false	0.7173	10.0	18.4 [14.1, 23.1]	52.8 [42.2, 64.5]	15.1 [11.0, 19.5]
DeepSets	top1	keep 90% covered	0.5067	86.3	87.5 [84.8, 90.0]	33.2 [28.7, 37.7]	86.1 [83.2, 88.8]
DeepSets	margin	at most 10% false	0.0636	10.0	8.9 [7.1, 10.8]	67.4 [58.1, 75.2]	9.9 [7.2, 13.1]
DeepSets	margin	keep 90% covered	0.0013	89.3	86.0 [81.7, 89.5]	32.8 [28.3, 37.1]	86.1 [81.8, 89.6]
Fusion (expl.)	top1	at most 10% false	2.9930	10.0	21.8 [18.0, 26.0]	86.8 [79.2, 93.1]	10.2 [7.7, 13.0]
Fusion (expl.)	top1	keep 90% covered	1.6547	86.8	86.9 [82.5, 90.9]	70.7 [65.8, 75.5]	83.8 [79.1, 88.0]
Fusion (expl.)	margin	at most 10% false	0.8766	10.0	14.0 [11.0, 17.6]	81.5 [72.8, 88.2]	9.9 [7.5, 12.7]
Fusion (expl.)	margin	keep 90% covered	0.0256	89.8	88.2 [84.2, 91.5]	66.4 [61.2, 71.3]	87.8 [84.0, 91.0]
Mass error	top1	at most 10% false	-0.0037	10.0	12.2 [9.9, 14.7]	22.1 [17.5, 27.4]	12.1 [9.8, 14.6]
Mass error	top1	keep 90% covered	-1.0102	90.2	90.0 [87.0, 92.7]	14.7 [12.3, 17.1]	90.0 [87.0, 92.7]
Mass error	margin	at most 10% false	0.1905	10.0	11.6 [8.0, 16.2]	4.9 [2.8, 8.6]	11.7 [8.1, 16.4]
Mass error	margin	keep 90% covered	0.0000	100.0	100.0 [100.0, 100.0]	13.3 [11.2, 15.6]	100.0 [100.0, 100.0]

Table 12: Descriptive. **Recall@5 (%) by instrument stratum**, test fold, deduplicated official pools, 95% intervals. Pre-specified as descriptive in the frozen protocol (§5) and labelled post hoc in the original article; 862 test spectra could not be mapped. Confounded by which molecules each instrument measured. Source: archived `analysis/strata-post-hoc.csv`.

Instrument	Spectra / molecules	Mass error	DeepSets	DreaMS + Morgan	Emb-Cos	Fusion
Orbitrap	7,643 / 1,071	13.5 [11.4, 16.1]	29.0 [24.6, 33.4]	57.2 [51.3, 63.1]	59.2 [53.9, 64.5]	61.2 [55.3, 66.9]
QTOF	2,143 / 489	13.9 [10.2, 17.9]	38.5 [32.8, 44.3]	71.2 [64.7, 76.8]	72.7 [67.5, 77.8]	73.0 [66.6, 78.5]
unmapped	862 / 302	10.1 [7.7, 12.7]	34.3 [28.4, 40.0]	74.1 [67.8, 80.1]	75.3 [70.3, 79.8]	76.3 [70.3, 82.0]

Table 13: Exploratory. **Fusion weight selection:** validation Recall@5 (%) on deduplicated official pools for each mass weight w ; $w = 0.55$ was chosen. Source: `companion/results/fusion-weight.json` (imported companion file).

w	Val. R@5	w	Val. R@5
0.00	64.74	0.55	68.26
0.05	64.97	0.60	68.14
0.10	65.25	0.65	67.54
0.15	65.66	0.70	67.08
0.20	66.09	0.75	66.10
0.25	66.48	0.80	65.66
0.30	66.89	0.85	64.60
0.35	67.27	0.90	62.81
0.40	67.71	0.95	61.93
0.45	67.99	1.00	14.60
0.50	68.16		

Table 14: NEW MEASUREMENT (data audit). **Precursor mass error against the annotated structure**, percentage of spectra below each threshold and median absolute error (ppm). Source: `archived/receipts/A02-precursor-error-audit-.../summary.json`.

Fold	Spectra	<0.01	<0.1	<1	<5	<10	Median
training	210,825	19.0	38.5	66.7	86.3	92.9	0.216
validation	9,631	20.5	41.5	71.7	90.8	94.6	0.146

Table 16: NEW MEASUREMENT. **All archived paired contrasts** (percentage points, 95% molecule-bootstrap intervals). Only the three rows marked as primary were pre-declared; all others are descriptive. With 16 candidates every method has Recall@20 = 100%, so those contrasts are exactly zero. Source: `archived/analysis/contrasts.csv`.

Contrast	Pool	Rule	k	Primary	Difference [95% CI]
C1	official, raw	expected	1		+29.50 [+24.58, +34.81]
C1	official, raw	expected	5		+45.03 [+39.14, +50.93]
C1	official, raw	expected	20		+47.98 [+42.37, +53.47]
C2	official, raw	expected	1		-2.88 [-7.63, +2.54]
C2	official, raw	expected	5		-2.51 [-5.75, +0.53]
C2	official, raw	expected	20		-0.03 [-2.93, +2.78]
C3	official, raw	expected	1		+9.75 [+6.96, +12.75]
C3	official, raw	expected	5		+16.00 [+11.65, +20.35]
C3	official, raw	expected	20		+18.69 [+13.22, +23.83]
C1	official, raw	strict	1		+31.82 [+26.79, +37.26]
C1	official, raw	strict	5		+50.47 [+44.40, +56.50]
C1	official, raw	strict	20		+60.68 [+54.48, +66.34]
C2	official, raw	strict	1		-2.74 [-7.45, +2.75]
C2	official, raw	strict	5		-2.55 [-5.80, +0.44]
C2	official, raw	strict	20		-0.15 [-3.02, +2.66]
C3	official, raw	strict	1		+12.31 [+9.52, +15.49]
C3	official, raw	strict	5		+21.62 [+17.25, +26.13]
C3	official, raw	strict	20		+30.94 [+25.45, +36.09]
C1	official, dedup.	expected	1		+29.55 [+24.65, +34.83]
C1	official, dedup.	expected	5	yes	+48.04 [+42.19, +53.59]
C1	official, dedup.	expected	20		+51.95 [+46.53, +57.01]
C2	official, dedup.	expected	1		-2.90 [-7.66, +2.52]
C2	official, dedup.	expected	5	yes	-1.85 [-5.19, +1.35]
C2	official, dedup.	expected	20		+1.50 [-1.36, +4.14]
C3	official, dedup.	expected	1		+9.74 [+6.94, +12.76]
C3	official, dedup.	expected	5	yes	+18.02 [+13.40, +22.49]

Contrast	Pool	Rule	k	Primary	Difference [95% CI]
C3	official, dedup.	expected	20		+21.91 [+16.08, +27.50]
C1	official, dedup.	strict	1		+31.82 [+26.79, +37.26]
C1	official, dedup.	strict	5		+54.15 [+48.23, +59.74]
C1	official, dedup.	strict	20		+64.16 [+58.43, +69.47]
C2	official, dedup.	strict	1		-2.74 [-7.45, +2.75]
C2	official, dedup.	strict	5		-1.91 [-5.26, +1.27]
C2	official, dedup.	strict	20		+1.25 [-1.63, +3.89]
C3	official, dedup.	strict	1		+12.31 [+9.52, +15.49]
C3	official, dedup.	strict	5		+23.94 [+19.41, +28.57]
C3	official, dedup.	strict	20		+34.54 [+29.06, +39.84]
C1	16	expected	1		+45.80 [+40.37, +51.37]
C1	16	expected	5		+31.95 [+27.29, +36.71]
C1	16	expected	20		+0.00 [+0.00, +0.00]
C2	16	expected	1		-0.19 [-3.71, +3.16]
C2	16	expected	5		+0.68 [-0.89, +2.11]
C2	16	expected	20		+0.00 [+0.00, +0.00]
C3	16	expected	1		+17.54 [+12.13, +23.12]
C3	16	expected	5		+11.37 [+5.65, +17.17]
C3	16	expected	20		+0.00 [+0.00, +0.00]
C1	16	strict	1		+55.40 [+49.79, +61.14]
C1	16	strict	5		+54.02 [+48.27, +59.67]
C1	16	strict	20		+0.00 [+0.00, +0.00]
C2	16	strict	1		-0.14 [-3.64, +3.20]
C2	16	strict	5		+0.69 [-0.88, +2.11]
C2	16	strict	20		+0.00 [+0.00, +0.00]
C3	16	strict	1		+26.68 [+21.33, +32.02]
C3	16	strict	5		+33.43 [+27.01, +40.10]
C3	16	strict	20		+0.00 [+0.00, +0.00]
C1	64	expected	1		+41.50 [+35.79, +47.23]
C1	64	expected	5		+50.73 [+44.94, +55.89]
C1	64	expected	20		+31.16 [+26.83, +35.79]
C2	64	expected	1		-1.79 [-5.40, +1.44]
C2	64	expected	5		+1.00 [-2.17, +4.06]
C2	64	expected	20		+1.44 [-0.09, +2.95]
C3	64	expected	1		+13.46 [+9.69, +17.33]
C3	64	expected	5		+21.15 [+15.20, +26.81]
C3	64	expected	20		+10.42 [+5.02, +16.01]
C1	64	strict	1		+45.96 [+40.13, +51.79]
C1	64	strict	5		+62.79 [+56.95, +67.95]
C1	64	strict	20		+52.52 [+47.15, +58.40]
C2	64	strict	1		-1.75 [-5.32, +1.49]
C2	64	strict	5		+0.71 [-2.52, +3.78]
C2	64	strict	20		+1.44 [-0.09, +2.93]
C3	64	strict	1		+18.13 [+14.42, +22.22]
C3	64	strict	5		+33.37 [+27.42, +39.09]
C3	64	strict	20		+31.78 [+25.45, +38.28]
C1	expanded	expected	1		+17.11 [+13.61, +20.76]
C1	expanded	expected	5		+33.15 [+27.64, +38.81]
C1	expanded	expected	20		+43.50 [+37.91, +48.90]
C2	expanded	expected	1		-4.26 [-8.07, -0.12]
C2	expanded	expected	5		-4.56 [-7.95, -1.19]
C2	expanded	expected	20		-3.06 [-7.50, +0.94]
C3	expanded	expected	1		+5.69 [+3.91, +7.55]
C3	expanded	expected	5		+11.73 [+8.39, +15.09]
C3	expanded	expected	20		+18.96 [+13.72, +24.20]
C1	expanded	strict	1		+18.00 [+14.47, +21.73]
C1	expanded	strict	5		+37.29 [+31.68, +43.05]
C1	expanded	strict	20		+49.87 [+43.82, +55.66]
C2	expanded	strict	1		-4.11 [-7.92, +0.02]
C2	expanded	strict	5		-4.61 [-8.02, -1.23]
C2	expanded	strict	20		-3.32 [-7.69, +0.66]
C3	expanded	strict	1		+6.95 [+5.11, +8.96]
C3	expanded	strict	5		+15.68 [+12.22, +19.22]
C3	expanded	strict	20		+25.79 [+20.55, +30.85]

Table 15: NEW MEASUREMENT, descriptive. **Pre-declared secondary population:** the official MassSpec-Gym `mces_1` test fold (17,556 spectra, 2,997 molecules), cheap baselines only, Recall@k (%) with 95% molecule-bootstrap intervals; random ordering is the analytic expectation. Under the strict rule every candidate ties for random ordering, so it scores 0 except where the pool is no larger than k (pool 16 at $k = 20$). Median deduplicated pool size 251 (`sub64` minimum 24). These results were archived but not reported in the original article. They are not comparable with Tables 4 and 10: the fold and molecules differ, and its share of recomputed-looking precursors was not audited. Source: archived `analysis-mces1/metrics.csv`.

Pool	Rule	Method	k	Recall [95% CI]
official, raw	expected	Mass error	1	7.6 [6.5, 8.8]
official, raw	expected	Mass error	5	22.5 [20.3, 24.8]
official, raw	expected	Mass error	20	46.8 [43.2, 50.4]
official, raw	expected	Random	1	0.4 [0.4, 0.4]
official, raw	expected	Random	5	2.0 [2.0, 2.0]
official, raw	expected	Random	20	8.0 [7.9, 8.1]
official, raw	strict	Mass error	1	3.0 [2.1, 4.2]
official, raw	strict	Mass error	5	11.1 [9.4, 12.9]
official, raw	strict	Mass error	20	30.9 [27.8, 34.2]
official, raw	strict	Random	1	0.0 [0.0, 0.0]
official, raw	strict	Random	5	0.0 [0.0, 0.0]
official, raw	strict	Random	20	0.0 [0.0, 0.0]
official, dedup.	expected	Mass error	1	7.7 [6.6, 9.0]
official, dedup.	expected	Mass error	5	23.4 [21.2, 25.7]
official, dedup.	expected	Mass error	20	48.8 [45.2, 52.4]
official, dedup.	expected	Random	1	0.4 [0.4, 0.4]
official, dedup.	expected	Random	5	2.1 [2.1, 2.2]
official, dedup.	expected	Random	20	8.6 [8.3, 8.9]
official, dedup.	strict	Mass error	1	3.0 [2.1, 4.2]
official, dedup.	strict	Mass error	5	11.4 [9.7, 13.3]
official, dedup.	strict	Mass error	20	32.7 [29.5, 36.1]
official, dedup.	strict	Random	1	0.0 [0.0, 0.0]
official, dedup.	strict	Random	5	0.0 [0.0, 0.0]
official, dedup.	strict	Random	20	0.0 [0.0, 0.0]
16	expected	Mass error	1	37.0 [33.9, 40.0]
16	expected	Mass error	5	74.6 [71.3, 77.8]
16	expected	Mass error	20	100.0 [100.0, 100.0]
16	expected	Random	1	6.2 [6.2, 6.2]
16	expected	Random	5	31.2 [31.2, 31.2]
16	expected	Random	20	100.0 [100.0, 100.0]
16	strict	Mass error	1	24.6 [21.8, 27.6]
16	strict	Mass error	5	64.7 [60.5, 68.5]
16	strict	Mass error	20	100.0 [100.0, 100.0]
16	strict	Random	1	0.0 [0.0, 0.0]
16	strict	Random	5	0.0 [0.0, 0.0]
16	strict	Random	20	100.0 [100.0, 100.0]
64	expected	Mass error	1	17.9 [16.0, 19.9]
64	expected	Mass error	5	45.9 [42.4, 49.4]
64	expected	Mass error	20	74.6 [70.9, 78.1]
64	expected	Random	1	1.6 [1.6, 1.6]
64	expected	Random	5	7.8 [7.8, 7.8]
64	expected	Random	20	31.3 [31.2, 31.3]
64	strict	Mass error	1	8.9 [7.4, 10.8]
64	strict	Mass error	5	31.5 [28.1, 34.9]
64	strict	Mass error	20	65.8 [61.5, 69.8]
64	strict	Random	1	0.0 [0.0, 0.0]
64	strict	Random	5	0.0 [0.0, 0.0]
64	strict	Random	20	0.0 [0.0, 0.0]

D Provenance, supplement and evidence mapping

Supplement S1 (original article). The full original article, *An MS/MS Shortlist Is Not an Identification: Choosing a Formula-Free Ranker and Knowing When to Decline* (5 October 2026), is retained unchanged as `article/published-original.md` (byte-identical to the published source) and is embedded in this PDF as the attachment `S1-published-original.md`; `article/original.md` differs only in local links. Every scientific result it reports appears in this manuscript, and `content-coverage.md` in `evidence/paper-migration/` maps it section by section.

Evidence ledger. `evidence/paper-migration/claims-ledger.json` links each substantive claim to the historical artifact (archive member, companion file or article text), a pointer inside it and its SHA-256. The results archive was read in memory without extraction; its member digests are recorded in `evidence/migration-audit.json`. The table generator (`scripts/paper_extract.py`) stops the build if any value that also appears in the original article differs from it at the printed precision; the recorded cross-check found no difference. Values that the original did not print (most appendix tables, including the `mces_1` secondary population) are formatted directly from the digest-checked archive members. A few numbers exist only as article text or in protocol notes (for example the repeated-structure counts, the near-duplicate share and the clean-archive CLI timings); the ledger marks them as such.

Citation note. Literature claims were verified for the original article (citation status “verified”) and were not re-retrieved during migration. Table 1 is transcribed from the original article.